



I2C – L3GD20HTR 테스트

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Mango-M32F2 I2C-L3GD20HTR 3- Axis Gyro sensor 테스트

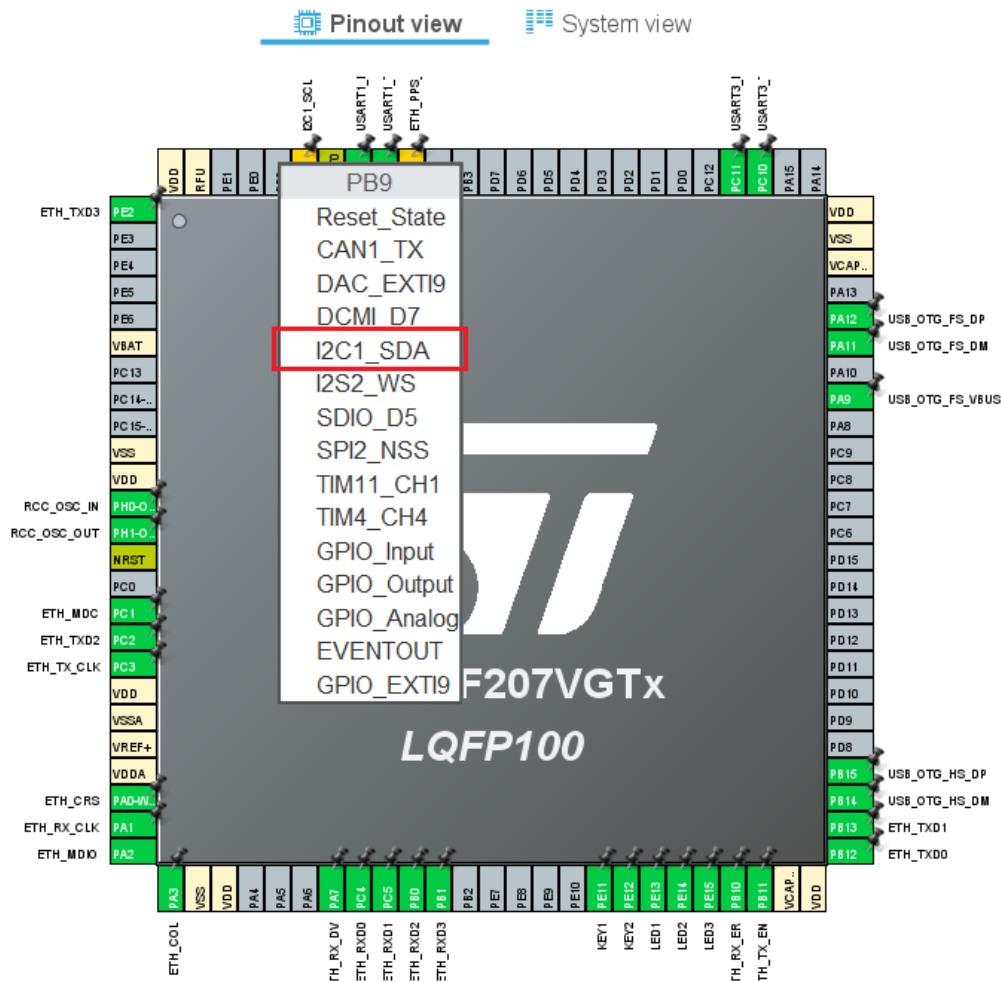
I2C pinmap

I2C – PB8, PB9

PB7	93	PB7	UART1_TX
PB8	95	PB8	I2C1_SCL
PB9	96	PB9	I2C1_SDA
PB10	47	PB10	MII_RX_FR

Pinout view

■ PB8-I2C1_SCL, PB9-I2C1_SDA 선택.



I2C 설정

- “Connectivity” 클릭, “I2C1” 클릭.
- “Mode” - “I2C” 클릭.

The screenshot displays the 'bma280_prj.ioc - Pinout & Configuration' window. The 'Pinout & Configuration' tab is active, showing a list of components on the left and configuration options on the right. In the left sidebar, under the 'Connectivity' category, 'I2C1' is selected and highlighted with a red box. The main panel shows the 'I2C1 Mode and Configuration' settings. The 'Mode' dropdown menu is set to 'I2C' and is also highlighted with a red box. Below this, the 'Configuration' section includes a 'Reset Configuration' button and tabs for 'NVIC Settings', 'DMA Settings', 'GPIO Settings', 'Parameter Settings' (which is active), and 'User Constants'. The 'Parameter Settings' tab shows the following configuration parameters:

Configure the below parameters :	
Search (Ctrl+F)	
Master Features	
I2C Speed Mode	Standard Mode
I2C Clock Speed (Hz)	100000
Slave Features	
Clock No Stretch Mode	Disabled
Primary Address Length	7 bit

L3GD20HTR Test Project

■ Code 생성 후 main.c 수정

```
36  /* Private define ----- */
37  /* USER CODE BEGIN PD */
38
39  #define GYRO_SLAVE_ADDRESS 0xD6
40
41  #define WHO_AM_I          0x0F
42  #define CTRL_REG1         0x20
43  #define CTRL_REG2         0x21
44  #define CTRL_REG3         0x22
45  #define CTRL_REG4         0x23
46  #define CTRL_REG5         0x24
47  #define HP_FILTER_RESET  0x25
48  #define STATUS_REG        0x27
49  #define OUT_X_L           0x28
50  #define OUT_X_H           0x29
51  #define OUT_Y_L           0x2A
52  #define OUT_Y_H           0x2B
53  #define OUT_Z_L           0x2C
54  #define OUT_Z_H           0x2D
55  #define INT1_CFG          0x30
56  #define INT1_SOURCE       0x31
57  #define INT1_THS          0x32
58  #define INT1_DURATION     0x33
59  #define INT2_CFG          0x34
60  #define INT2_SOURCE       0x35
61  #define INT2_THS          0x36
62  #define INT2_DURATION     0x37
63
64  typedef enum adc_data
65  {
66      X_VALUE=0,
67      Y_VALUE,
68      Z_VALUE,
69
70      NUM_OF_VALUE
71  } ADC_DATA;
72
73  /* USER CODE END PD */
74
```

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```
102 /* Private user code -----*/
103 /* USER CODE BEGIN 0 */
104 static unsigned int count = 0;
105
106 static uint8_t Read_1_Byte(uint8_t address)
107 {
108     uint8_t Buffer_Tx;
109     uint8_t Buffer_Rx;
110     HAL_StatusTypeDef i2cRW_status;
111
112     Buffer_Tx = address;
113
114     i2cRW_status = HAL_I2C_Master_Transmit(&hi2c1, GYRO_SLAVE_ADDRESS, &Buffer_Tx, 1, 1000);
115     if(HAL_OK != i2cRW_status)
116     {
117         printf("Write error:%d\r\n", i2cRW_status);
118         return 0;
119     }
120
121     i2cRW_status = HAL_I2C_Master_Receive(&hi2c1, GYRO_SLAVE_ADDRESS, &Buffer_Rx, 1, 1000);
122     if(HAL_OK != i2cRW_status)
123     {
124         printf("Read error:%d\r\n", i2cRW_status);
125         return 0;
126     }
127
128     return Buffer_Rx;
129 }
130
131 static uint8_t Get_ChipID(void)
132 {
133     return Read_1_Byte(WHO_AM_I);
134 }
```

L3GD20HTR Test Project

```
165 static void Write_1_Byte(uint8_t address, uint8_t data)
166 {
167     uint8_t Buffer_Tx[2];
168     HAL_StatusTypeDef i2cRW_status;
169
170     Buffer_Tx[0] = address;
171     Buffer_Tx[1] = data;
172
173     i2cRW_status = HAL_I2C_Master_Transmit(&hi2c1, GYRO_SLAVE_ADDRESS, Buffer_Tx, 2, 1000);
174     if(HAL_OK != i2cRW_status)
175     {
176         printf("Write error:%d\r\n", i2cRW_status);
177     }
178 }
179
180 /*
181  * Description : none.
182  * Argument(s) : none.
183  * Return(s)    : none.
184  * Note(s)      : none.
185  */
186 static void GYRO_L3G4200_vInit (void)
187 {
188     // Enable x, y, z and turn off power down:
189     Write_1_Byte(CTRL_REG1, 0x0F);
190
191     // If you'd like to adjust/use the HPF, you can edit the line below to configure CTRL_REG2:
192     Write_1_Byte(CTRL_REG2, 0x0);
193
194     // Configure CTRL_REG3 to generate data ready interrupt on INT2
195     // No interrupts used on INT1, if you'd like to configure INT1
196     // or INT2 otherwise, consult the datasheet:
197     Write_1_Byte(CTRL_REG3, 0x08);
198
199     // CTRL_REG5 controls high-pass filtering of outputs, use it
200     // if you'd like:
201     Write_1_Byte(CTRL_REG5, 0x0);
202 }
203 /* USER CODE END 0 */
```


L3GD20HTR Test Project

```
209 int main(void)
210 {
211     /* USER CODE BEGIN 1 */
212     u8_t u8X_H, u8X_L, u8Y_H, u8Y_L, u8Z_H, u8Z_L;
213     signed short n16Value[NUM_OF_VALUE]={0};
214     /* USER CODE END 1 */
```

```
241     /* USER CODE BEGIN 2 */
242     printf("I2C-L3GD20HTR test\r\n");
243     httpd_init();
244
245     GYRO_L3G4200_vInit();
246     printf("Get_ChipID: 0x%X\r\n", Get_ChipID());
247     /* USER CODE END 2 */
```

L3GD20HTR Test Project

```
249  /* Infinite loop */
250  /* USER CODE BEGIN WHILE */
251  while (1)
252  {
253      MX_LWIP_Process();
254
255      if(count++ % 10000 == 0)
256      {
257          // ROLL
258          u8X_H = Read_1_Byte(OUT_X_H);
259          u8X_L = Read_1_Byte(OUT_X_L);
260
261          // PITCH
262          u8Y_H = Read_1_Byte(OUT_Y_H);
263          u8Y_L = Read_1_Byte(OUT_Y_L);
264
265          // YAW
266          u8Z_H = Read_1_Byte(OUT_Z_H);
267          u8Z_L = Read_1_Byte(OUT_Z_L);
268
269          n16Value[X_VALUE] = (signed short)((u8X_H << 8) | u8X_L);
270          n16Value[Y_VALUE] = (signed short)((u8Y_H << 8) | u8Y_L);
271          n16Value[Z_VALUE] = (signed short)((u8Z_H << 8) | u8Z_L);
272
273          printf ("X: %d, Y: %d, Z: %d\r\n", n16Value[X_VALUE], n16Value[Y_VALUE], n16Value[Z_VALUE]);
274      }
275  /* USER CODE END WHILE */
```

Test 결과 확인

- CR-zF-L3GD20HTR 센서 보드를 8Pin Cable을 이용 CON7에 연결함.
- Chip ID – 0xD7 정상적으로 읽힘.
- 센서를 움직일 때 3-axis 데이터가 변동함.

```
COM4 - Tera Term VT
File Edit Setup Control Window Help
I2C-L3GD20HTR test
Get_ChipID: 0xD7
X: 155, Y: -132, Z: -52
X: -3459, Y: 5268, Z: 4295
X: -206, Y: 740, Z: 161
X: 9, Y: 699, Z: 512
X: 13, Y: 395, Z: 636
X: -65, Y: 665, Z: 481
X: -18, Y: 605, Z: 492
X: 626, Y: 1160, Z: 1315
X: -387, Y: -601, Z: 213
X: 117, Y: 1425, Z: 459
X: -126, Y: 2, Z: 574
X: 103, Y: 978, Z: 311
X: -223, Y: -1008, Z: 429
X: 95, Y: 1715, Z: -188
X: 522, Y: 94, Z: 1081
X: 220, Y: 1255, Z: 572
X: -777, Y: -842, Z: -218
X: -82, Y: 2188, Z: 242
X: 121, Y: -92, Z: 1124
X: 1189, Y: 3586, Z: 1105
X: -1071, Y: -1944, Z: -648
X: -63, Y: 1683, Z: 461
X: -185, Y: 485, Z: 742
X: -207, Y: 872, Z: 255
X: 124, Y: 1417, Z: 481
X: -356, Y: 374, Z: 645
X: -164, Y: 968, Z: 669
X: 127, Y: 1663, Z: 326
```

Thank You

